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Travelling Salesperson Problem (TSP)

Video: Introduction to TSP and its applications

16 min

Video: NP-Hardness of TSPs

11 min

Video: Hardness of Approximating General TSPs

18 min

Lab: Interactive Notes on TSP Basics, NP-Hardness and Inapproximability

1h

Quiz: TSP Basics

30 min

Exact Algorithms for TSP

Approximation Algorithm for TSP

Heuristics: A Brief Tour

Approximation Schemes and the Knapsack Problem

Problem Set # 4

TSP Basics

Review Learning Objectives

Submit your assignment

Due Mar 3, 11:59 PM IST

Receive grade

To Pass 80% or higher

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Grade received 95%

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To pass 80% or higher

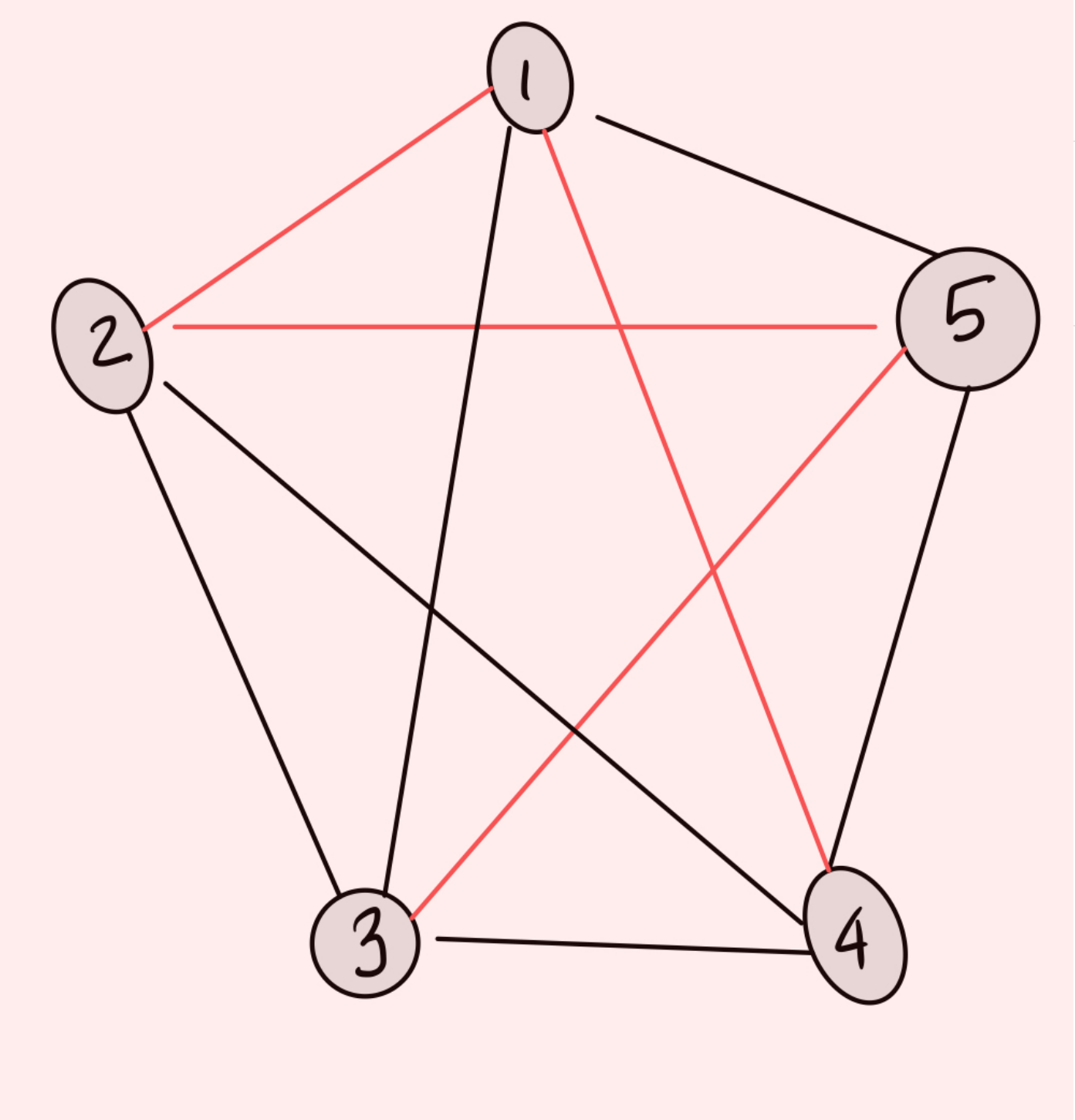
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Try again

Your grade 95%

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The questions below concern the following TSP instance involving 5 nodes:



Edges shown in black are assigned a weight of 1 whereas edges shown in red are assigned a weight of 2.

1. Select all the correct answers from the list below.

3 / 3 points

☒ There is a TSP tour whose cost is 5, i.e, it involves only the black edges in the graph.

☒ Correct

Correct: 1, 5, 4, 2, 3, 1 will do it.

☒ The sequence of vertices 1, 2, 3, 4, 5, 1 (1 is the start/end point) is a valid TSP tour of cost 6

☒ Correct

☐ The sequence of vertices 1, 2, 3, 4, 5, 1 (1 is the start/end point) is an optimal cost TSP tour.

☐ The sequence of vertices 1, 2, 3, 1, 5, 4, back to 1 is a valid TSP tour.
2. The TSP above is an instance of a Metric TSP. It is symmetric and satisfies the triangle inequality.

1 / 1 point

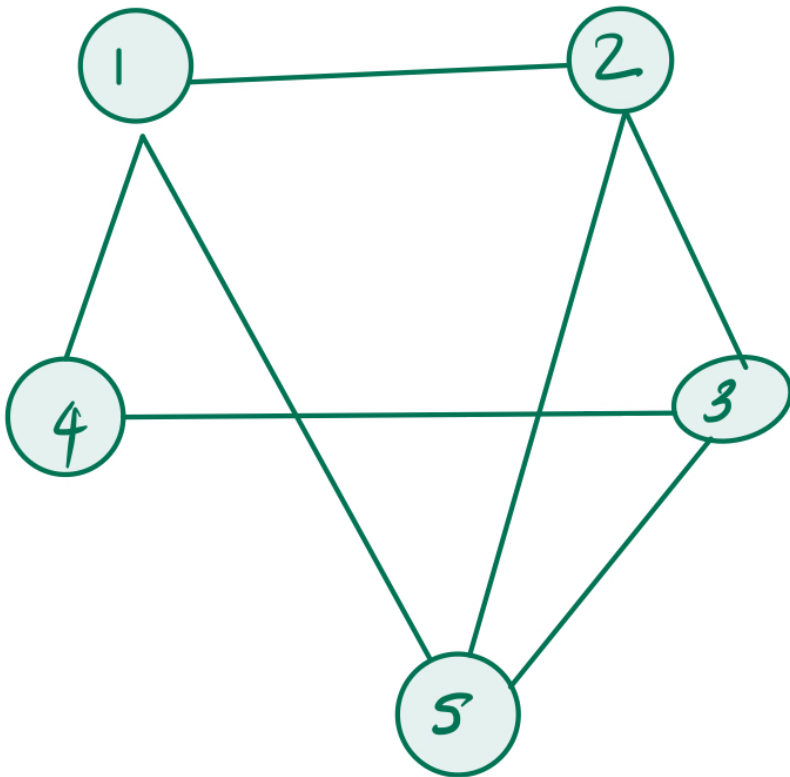
☐ False

☒ True

☒ Correct

Notice that every triple satisfies the triangle inequality
3. This question concerns the following instance of the Hamiltonian Cycle Problem:

0.75 / 1 point



- We wish to find out if there is a Hamiltonian cycle problem using a reduction to the TSP problem. Select all the correct answers about this reduction.
- ☐ Once we reduce to the TSP, we conclude the presence of a Hamiltonian cycle if the optimal TSP tour has weight 6.

☐ Lack of a Hamiltonian cycle will also yield an optimal TSP tour of weight 6.

☒ The reduction will need to assign a weight of 1 to each edge in the original graph and add the missing edges (4,5), (1,3) and (2,4) with some weight $W > 1$

☒ Correct

Correct.

☒ Let $W > 1$ be the weight given to the missing edges that we will need to add back for the reduction to TSP. The optimal TSP tour cost will be at least $5 + W$ if there is no Hamiltonian cycle.

☒ Correct

Correct.

You didn't select all the correct answers
-